Step-by-step guide

DVR and D-STATCOM with hybrid PV–wind generation on the IEEE 13-bus feeder

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This guide runs the whole study by hand in Simulink, from the plain feeder to the combined system. No script is used. Each step says which blocks to change, what to look at, and what every figure means.

The only file needed: `IEEE13_DVR_DSTATCOM_PV_Wind.slx`

> **How to use this guide.** Open the model once and keep it open. Go through the steps in order. Each step lists only the blocks that change from the step before. Do not press Save. To go back to the saved model, close it without saving.

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Getting started

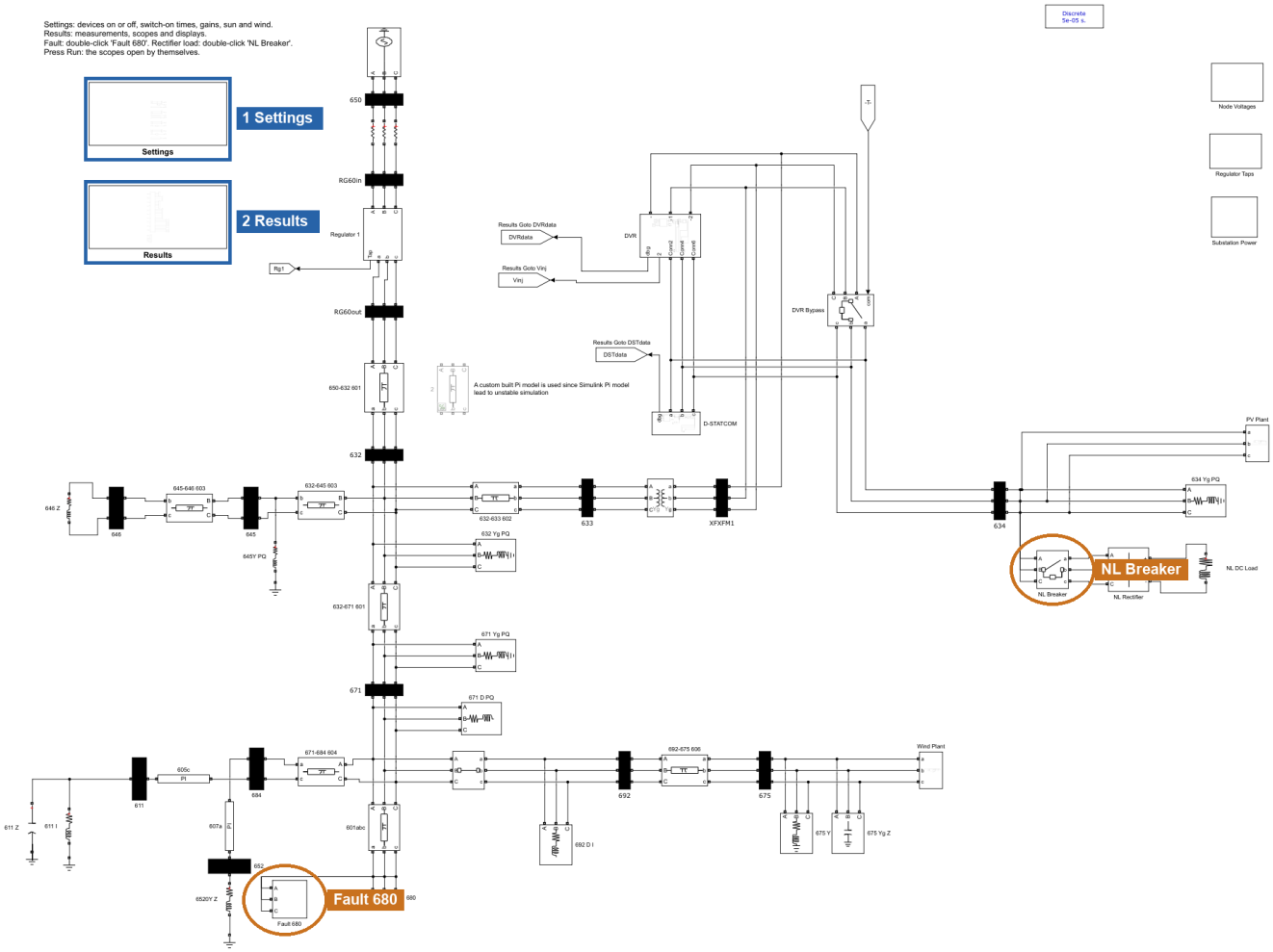
What you need

- MATLAB **R2024b** with **Simulink** and **Simscape Electrical** (Specialized Power Systems).
- For MATLAB R2022a or R2022b, use the copy in `MATLAB_R2022/Simulink_model/`.

The three places in the model

The main page shows the feeder and two subsystems. Double-click a subsystem to open it. To come back, click the model name in the bar above the diagram.

Where	What is there
1 Settings	A note and the settings of the DVR, D-STATCOM, PV plant and wind plant.
2 Results	The Measurements block, five scopes and the displays.
Main page	The IEEE 13-bus feeder with the four devices. **Fault 680** and **NL Breaker** are here.



The main page. Blue: the two subsystems. Orange: the two blocks in the feeder that you may change.

How to do the basic actions

To do this	Do this
Change a value	Double-click the block, edit the field, click **OK**.
Run the model	Press **Run** on the Simulation tab. A run takes about a minute. The scopes open by themselves.
Read the displays	After the run, double-click **Results**. They are on the right of the page.
Look closely at a part of a scope	Use the zoom tool in the scope window and drag over the part you want.
Save a figure	Press **Win + Shift + S** and capture the scope window.
Compare two runs	Simulation tab, then **Data Inspector**. Each run is listed. Tick the same signal in two runs to draw them together.
Go back to the saved model	Close the model without saving, then open it again.

Words used in this guide

Word	Meaning
pu (per unit)	A value divided by its normal value. 1.00 pu is normal voltage, and 0.60 pu is 60 % of normal.
RMS voltage	The effective size of an AC voltage. It is the easiest way to see a voltage change.
Sag	The voltage falls below 0.9 pu for a short time.
Swell	The voltage rises above 1.1 pu for a short time.
THD	Total harmonic distortion: how far a waveform is from a pure sine wave, in percent. 0 % is a pure sine wave.
Power factor	The share of the supplied power that is useful active power. 1.00 is the best.
Unbalance	How unequal the three phases are, in percent. 0 % means they are equal.
Active power (kW)	The power that does real work.
Reactive power (kvar)	The power that moves back and forth for motors and magnetic fields. It loads the lines but does no work.
LG, LL, LLG, LLLG	Fault types: one line to ground, line to line, two lines to ground, all three lines to ground.

Where each signal is measured

These names are important, because they do not always mean what they seem to mean.

Name in the scopes	What it really is
Supply voltage	Voltage before the DVR, at the transformer XFXFM1.
Load voltage	Voltage at node 634, after the DVR. This is the voltage the 480 V load receives.
DVR voltage	The voltage the DVR adds in series.
Node 675 voltage	Voltage at node 675 on the 4.16 kV side, close to the fault. The wind plant connects here.
Source current	Current coming from the transformer, before the D-STATCOM joins the line.
Load current	Current going into node 634. It is the load current minus what the PV plant supplies.
Power from source	Power reaching node 634 from the transformer. It is not the power of the whole feeder.

The five scopes and the displays

Scope	Panels	Use it to see
Voltage waveforms	Supply, DVR, Load voltage	The actual waves, and how the DVR adds to the supply
RMS voltages	Supply, Load, Node 675	Sags and swells. This is the most useful scope
Power quality	Voltage THD, current THD, power factor, current unbalance, voltage unbalance	How clean and balanced the supply is
Power and devices	DVR power and energy, D-STATCOM DC voltage, PV and wind power, active and reactive power from the source	What each device is doing
Currents	Source, Load, D-STATCOM, PV, Wind	The current in each branch

Displays (inside Results): lowest and highest load voltage, lowest supply voltage, peak DVR power, DVR energy, and the power factor, current THD, current unbalance and load voltage THD at the end of the run. The lowest and highest values are counted from 0.5 s, so the start of the run is not included.

Reading the figures in this guide

- They come from your model, with the same signals and the same axis limits as the scopes. Line colours may differ from your screen.
- The scopes stack their panels in one column. Here they are arranged in two columns to save space.
- Dotted vertical lines mark events such as “fault on”.
- Ignore the first 0.03 s of every plot. RMS, THD and unbalance need one full cycle before they read correctly.
- In the power quality and power plots, a line with one colour is one value for all three phases.

Other blocks on the main page

- **Substation Power**: displays of the power taken at the substation. Double-click to open.
- **Regulator Taps**: scopes of the voltage regulator tap positions.
- **Node Voltages**: shows nothing in this model, because its output block was removed when the model was built. It does not affect any result. Nodes 634 and 675 are read from Results.

Step 1: literature review

Step 1 is reading, not simulation. Ten papers were reviewed to find the research gap, so the work in Simulink starts at Step 2.

Step 2: only the IEEE 13-bus feeder

Aim. Show that the feeder alone gives the correct voltages before anything is added to it.

Change these blocks

Block	Where	Field	Set to
DVR in service (1 = yes, 0 = no)	Settings	Constant value	0
D-STATCOM in service (1 = yes, 0 = no)	Settings	Constant value	0
PV plant in service (1 = yes, 0 = no)	Settings	Constant value	0
Wind plant in service (1 = yes, 0 = no)	Settings	Constant value	0
Fault 680	Main page	Switching times (s)	[100 100.1]
NL Breaker	Main page	Switching times (s)	100

The stop time is 1 s, so a time of 100 means “never”. Any time after 1 s works, such as [5 10]. The devices stay on the drawing but are disconnected: their breakers stay open, and a bypass breaker closes around the DVR. Press **Run**.

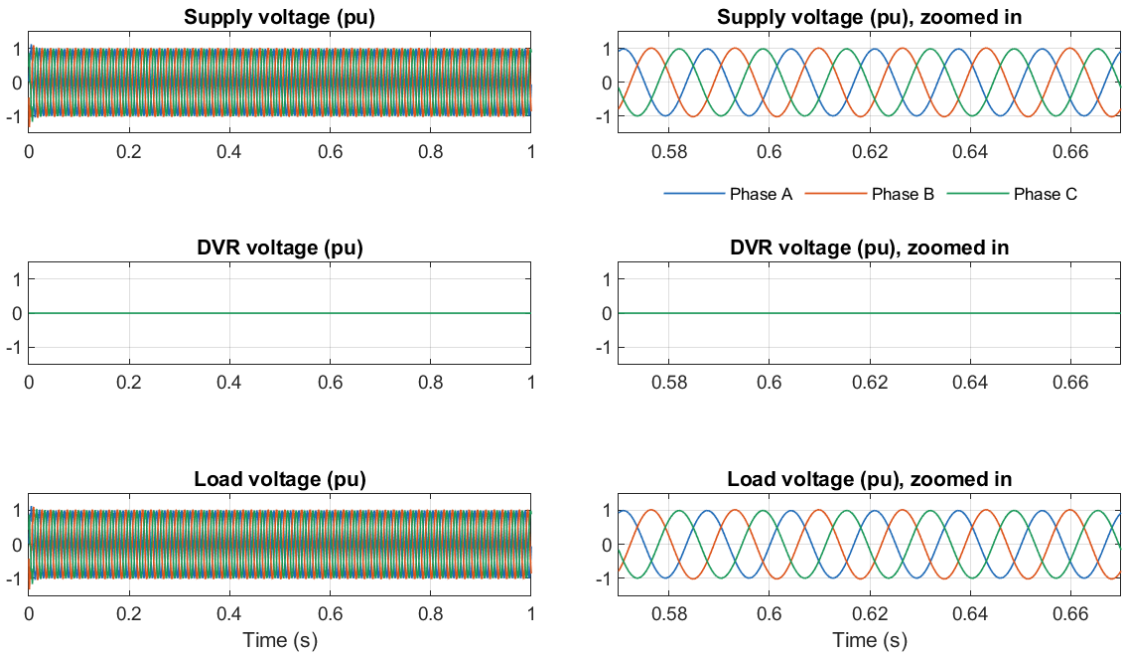
Check your result

Node	Source	Phase A	Phase B	Phase C
634	Your model	0.993	1.019	0.995
634	IEEE published	0.994	1.022	0.996
675	Your model	0.986	1.046	0.978
675	IEEE published	0.984	1.055	0.976

Result. Every value is within 0.01 pu of the published value, so the feeder is valid. The published values here should be checked against the IEEE 13-node test feeder document before they are quoted.

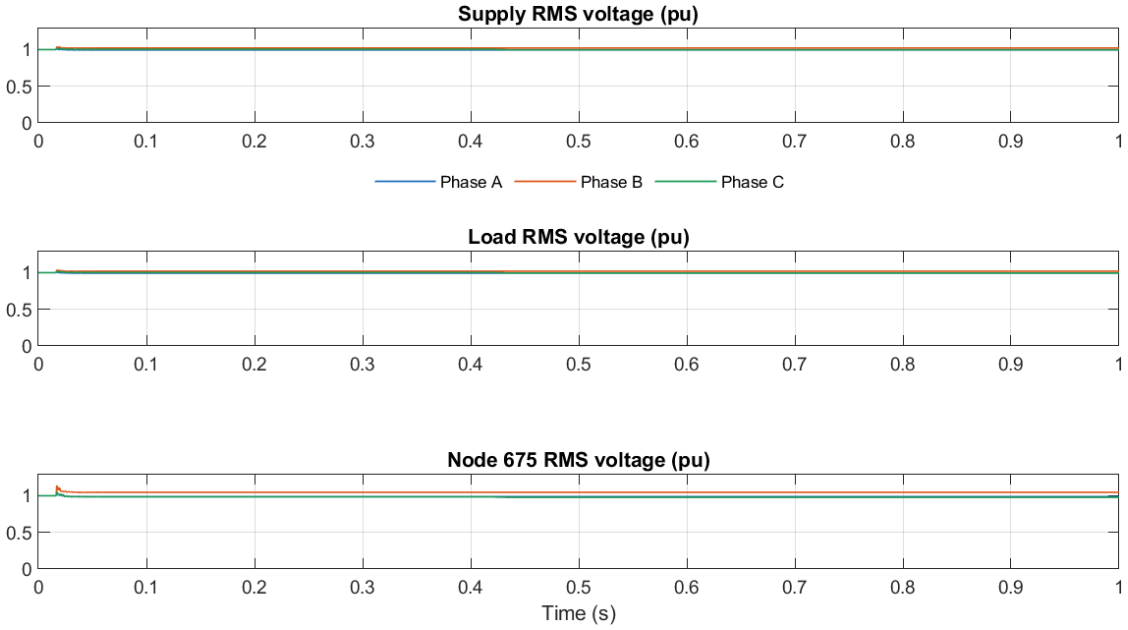
No fault or device acts in this step, so its figures have no dotted lines.

Figure 2.1 Voltage waveforms



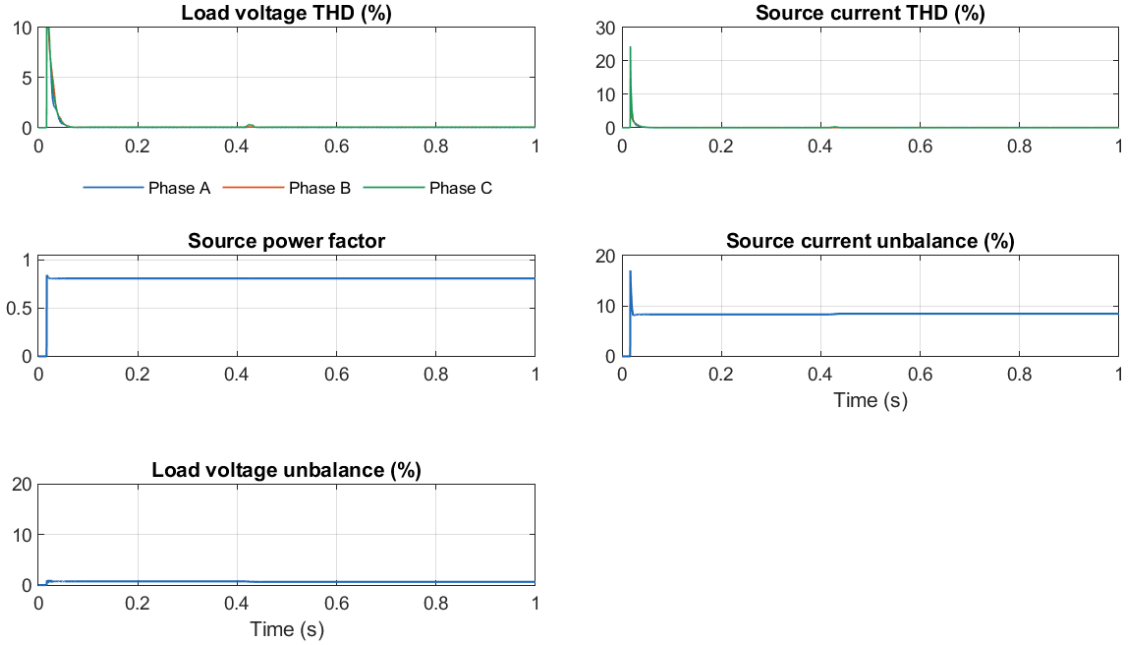
- Supply and load voltage are clean sine waves with the three phases evenly spaced. The right side zooms in to show this.
- The DVR voltage is zero, because the DVR is off.

Figure 2.2 RMS voltages



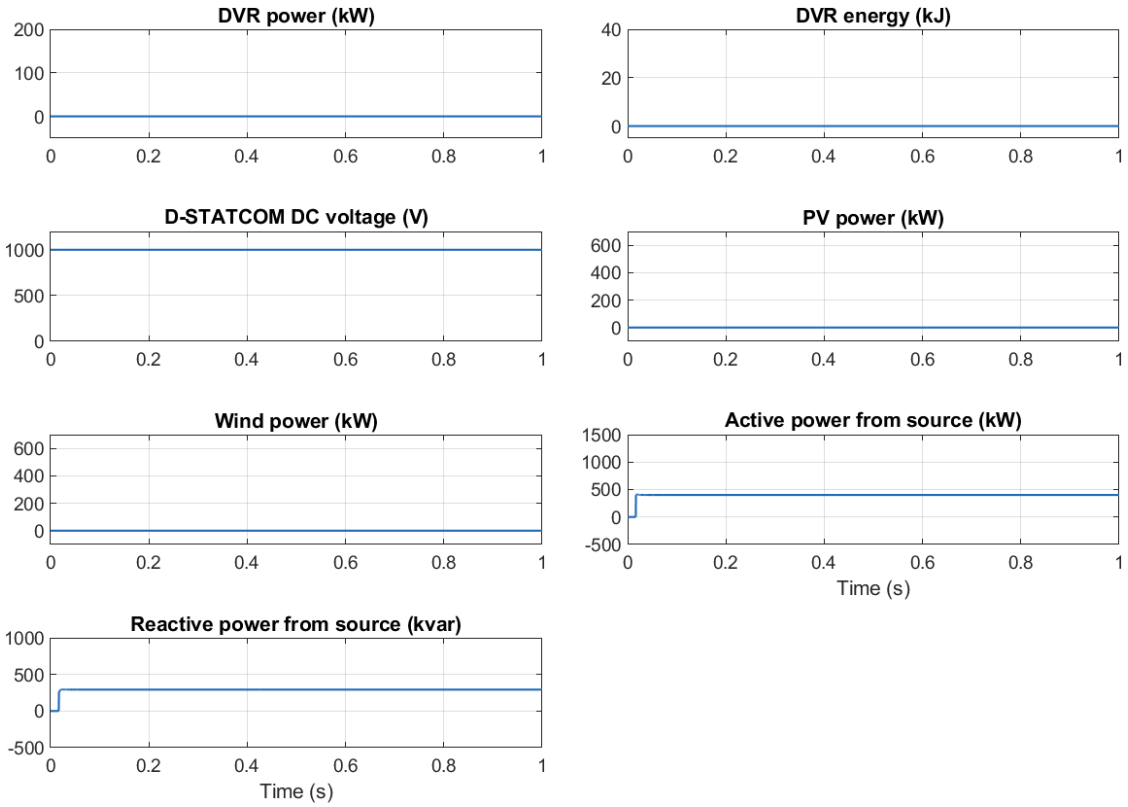
- All lines are flat: no sag and no swell. Supply is 0.995 / 1.020 / 0.996 pu and load is 0.993 / 1.019 / 0.995 pu.
- The phases differ slightly because the IEEE 13-bus feeder is unbalanced by design.
- Phase C falls slightly around 0.42 s, from 1.002 to 0.995 pu. No fault or device is active, so the voltage regulator changing tap is the only thing that can cause it.

Figure 2.3 Power quality



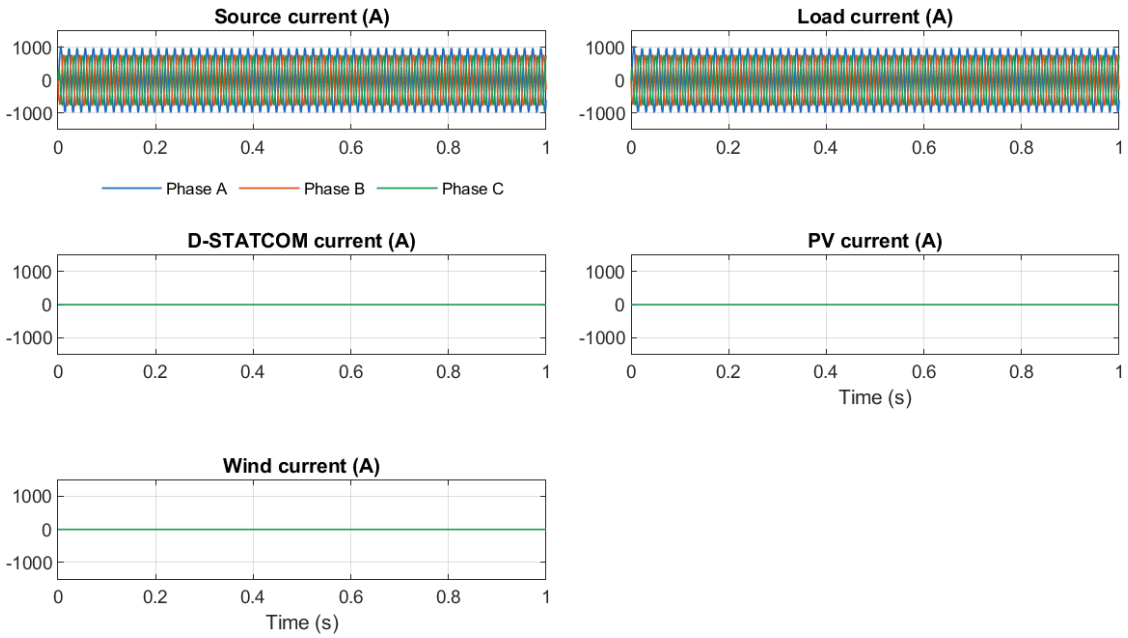
- THD is almost 0 %, because no rectifier is connected.
- The power factor is 0.81. The load at node 634 uses a lot of reactive power.
- Current unbalance is 8.4 % because the phases carry unequal loads. Voltage unbalance is only 0.6 %.

Figure 2.4 Power and devices



- The source gives 401 kW and 291 kvar, which matches the node 634 load of 400 kW and 290 kvar.
- DVR, PV and wind are all zero because they are off.
- The D-STATCOM DC voltage stays at 1000 V. This is only its starting value; it is disconnected.

Figure 2.5 Currents



- Source and load current are the same, because nothing is connected between them. Peaks are 983 / 780 / 761 A.
- Phase A carries the most current. D-STATCOM, PV and wind currents are zero.

Step 3: base case, with no compensation

Aim. See the power quality problem that the devices must solve: a fault and a non-linear load.

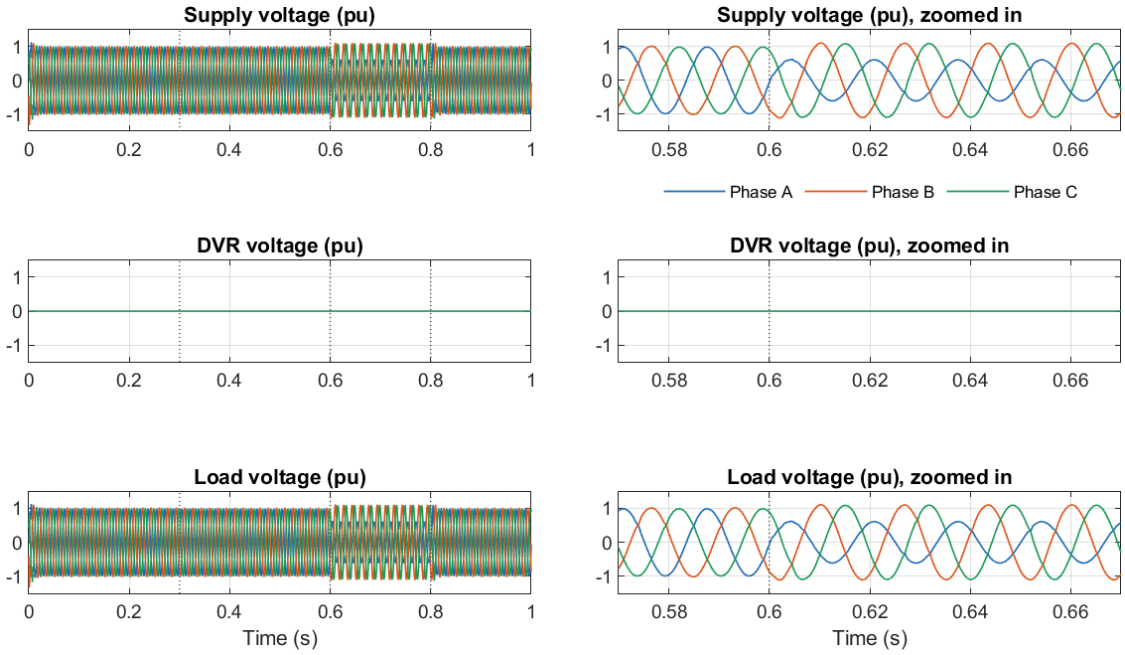
Change these blocks

Block	Where	Field	Set to
Fault 680	Main page	Switching times (s)	[0.6 0.8]
Fault 680	Main page	Tick boxes	Phase A and Ground ticked, others not
NL Breaker	Main page	Switching times (s)	0.3

All four devices stay at 0. Press **Run**. Dotted lines in the figures: rectifier on at 0.3 s, fault from 0.6 s to 0.8 s.

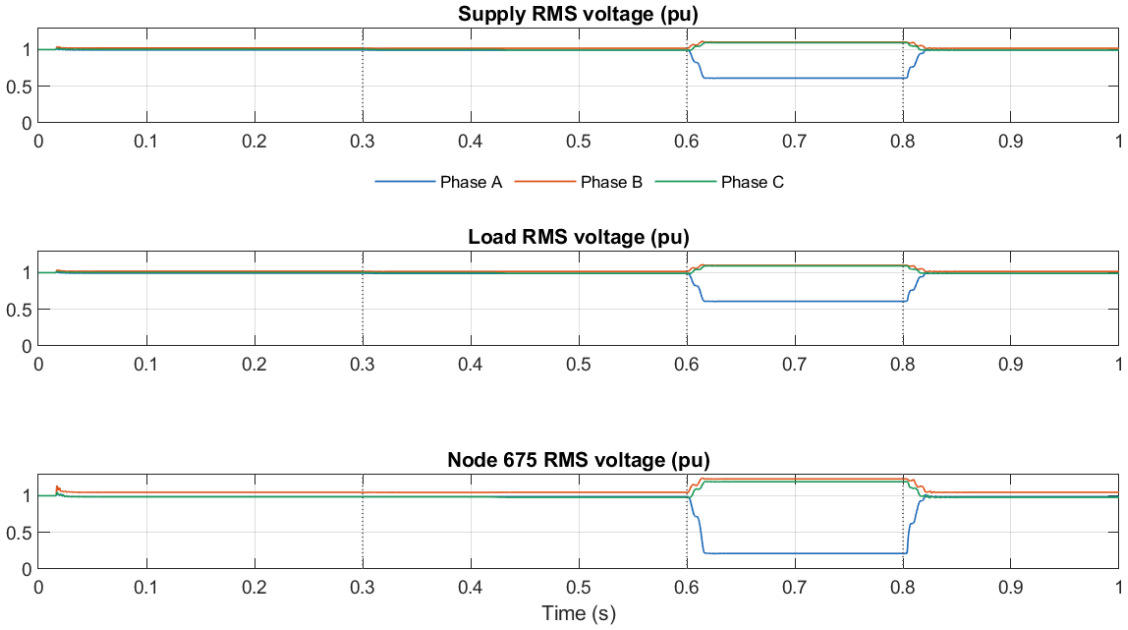
Result. The load sags to 0.61 pu on the faulted phase and swells to 1.10 pu on the other two, both at once. The rectifier adds harmonics. Nothing corrects these problems.

Figure 3.1 Voltage waveforms



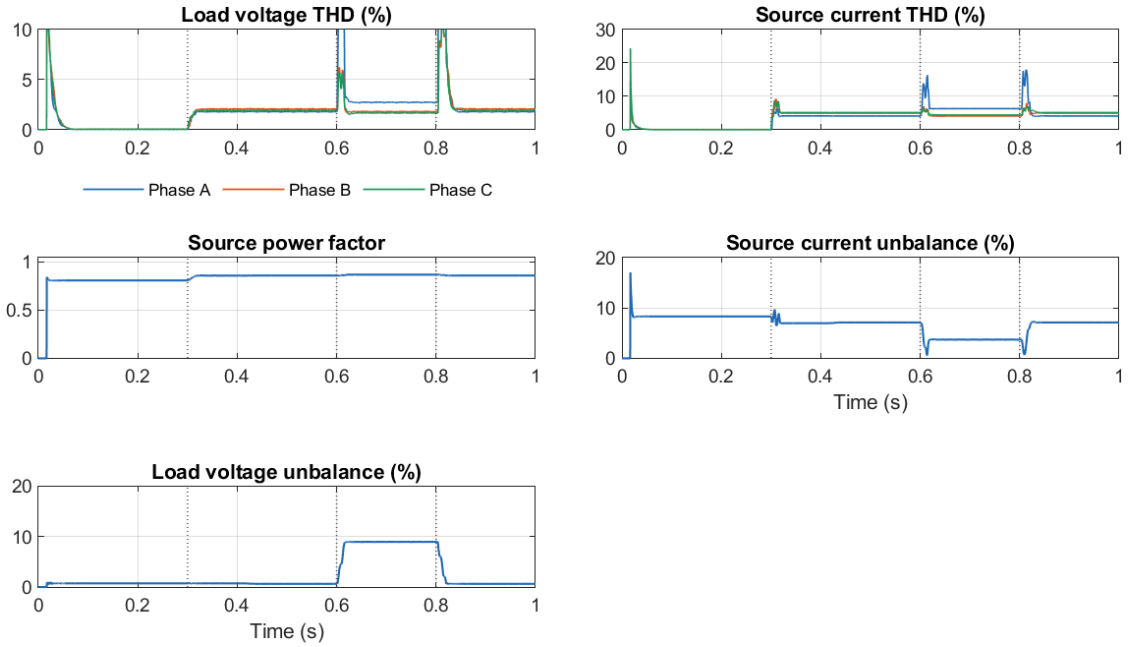
- At 0.6 s phase A becomes small while phases B and C become larger. The zoom shows this clearly.
- Load voltage has the same shape as supply voltage, because nothing corrects it. DVR voltage is zero.

Figure 3.2 RMS voltages



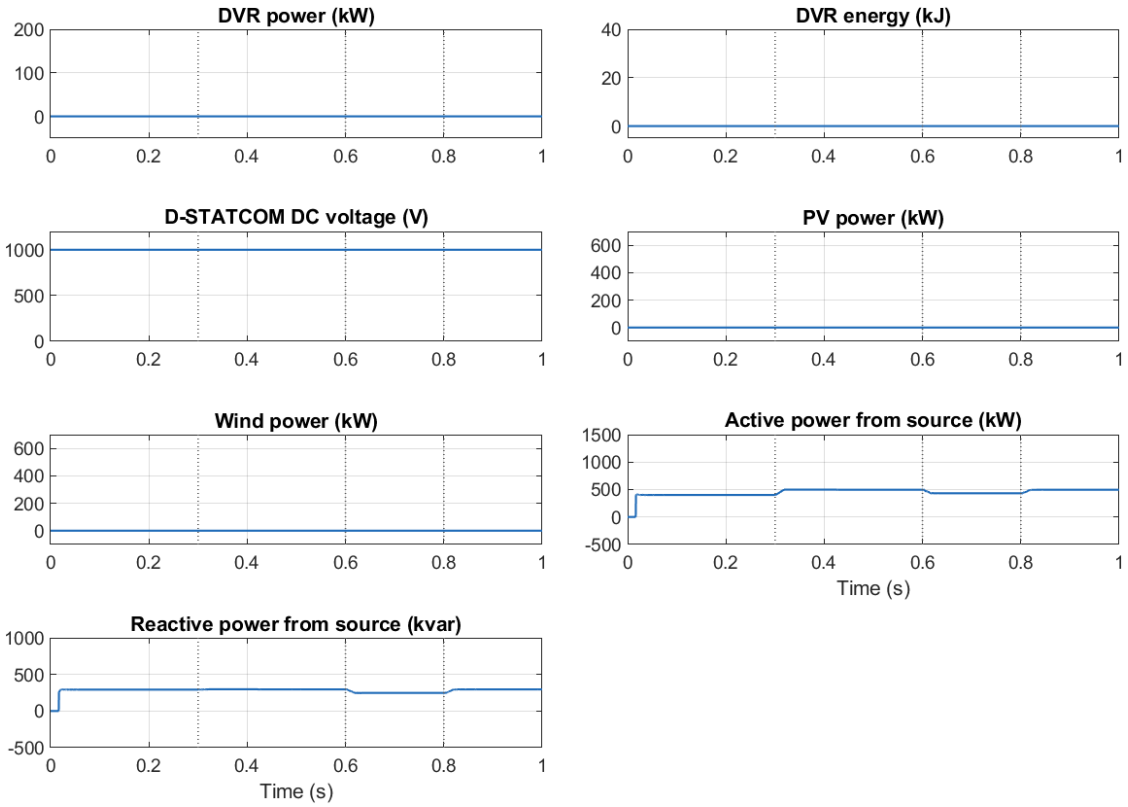
- Before the fault the load is 0.989 / 1.015 / 0.991 pu. During the fault it is 0.609 / 1.100 / 1.091 pu.
- Node 675 is closer to the fault, so it falls much further, to 0.211 / 1.228 / 1.190 pu.
- The voltage returns to normal when the fault ends at 0.8 s.

Figure 3.3 Power quality



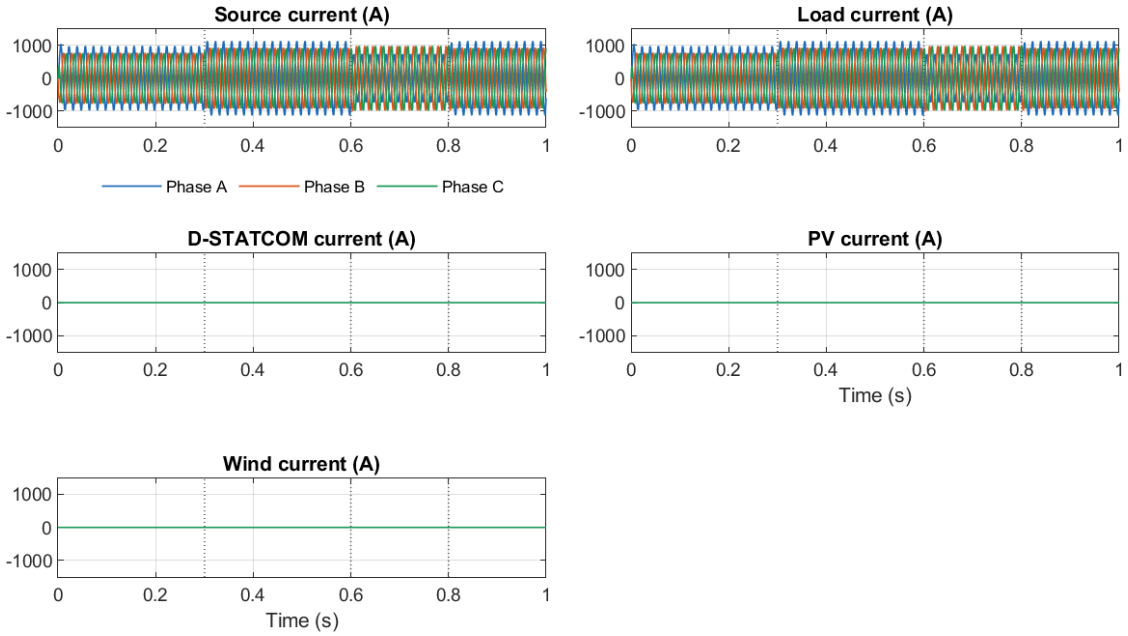
- At 0.3 s the rectifier connects. Voltage THD rises from about 0 to about 2 %, and current THD to 4–5 %.
- The power factor rises from 0.81 to 0.86, because the rectifier adds mainly active power.
- During the fault the load voltage unbalance jumps to 8.9 %.

Figure 3.4 Power and devices



- At 0.3 s active power rises from 403 to 497 kW: this is the rectifier load. Reactive power stays near 295 kvar.
- During the fault active power falls to 433 kW and reactive power to 247 kvar, because the load receives a different voltage.

Figure 3.5 Currents



- At 0.3 s the current grows. Before the fault the peaks are 1133 / 932 / 915 A.
- During the fault the phase A current falls to 738 A, and phases B and C rise to about 985 A.

Try the other fault types

Open **Fault 680**, change the tick boxes, and run again. Read the RMS voltages scope during the fault.

Fault	Tick	Load at node 634 (pu)	Node 675 (pu)
LLLG, three-phase to ground	A, B, C, Ground	0.604 / 0.631 / 0.617	0.208 / 0.220 / 0.212
LG, phase A to ground	A, Ground	0.609 / 1.100 / 1.091	0.211 / 1.228 / 1.190
LL, phases B and C	B, C	0.977 / 0.727 / 0.736	0.969 / 0.556 / 0.539
LLG, phases B and C to ground	B, C, Ground	1.096 / 0.615 / 0.614	1.211 / 0.220 / 0.210

Set the tick boxes back to **Phase A and Ground** before Step 4.

Step 4: D-STATCOM alone

Aim. See what a shunt device can do. The D-STATCOM injects current at node 634, in parallel with the load.

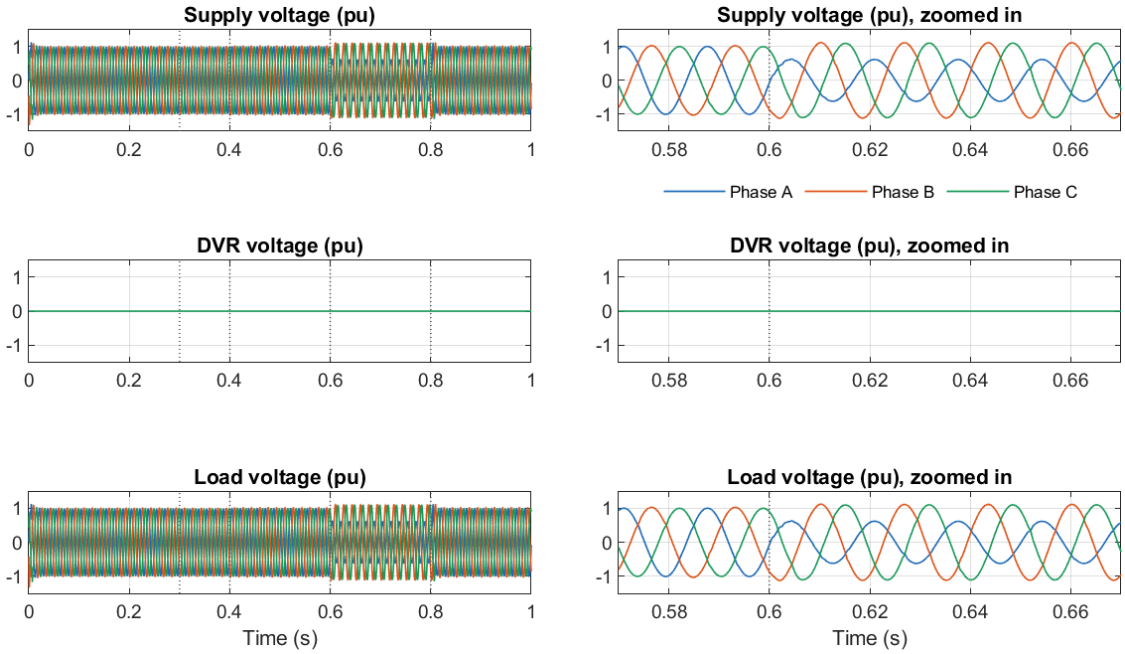
Change these blocks (inside Settings)

Block	Field	Set to
D-STATCOM in service (1 = yes, 0 = no)	Constant value	1
D-STATCOM gains [bandwidth (Hz); KpDc; KiDc; KpV; KiV; Imax (A)]	Constant value	[1500;1;15;5;500;850]

These are the hand-tuned gains. Keep the mode at 1 (load compensation). Press **Run**. Dotted lines: rectifier on at 0.3 s, D-STATCOM on at 0.4 s, fault from 0.6 s to 0.8 s.

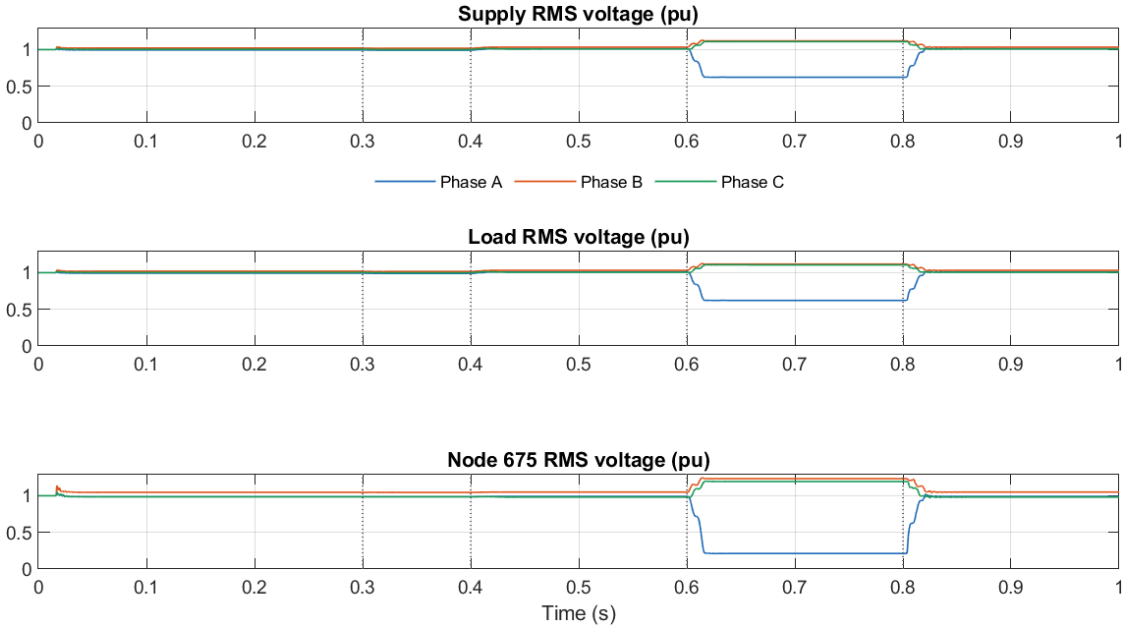
Result. The D-STATCOM cleans up the current: power factor 1.00, current unbalance 0.7 %, current THD 2.8 %. But it cannot repair the sag: the load still falls to about 0.62 pu.

Figure 4.1 Voltage waveforms



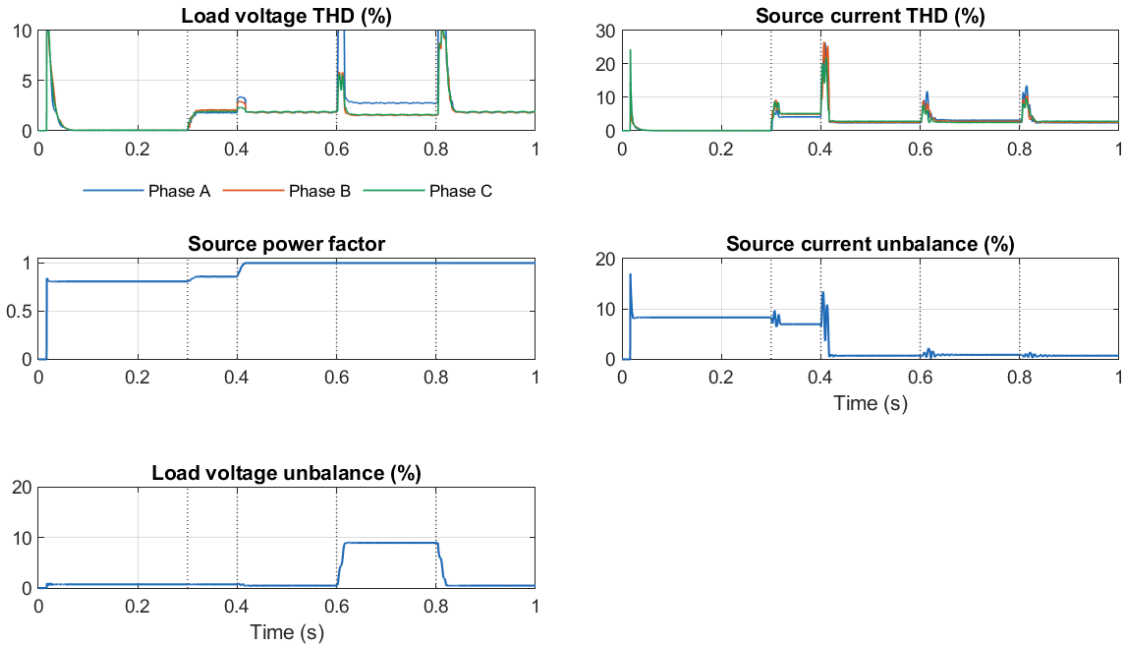
- The fault looks almost the same as in Step 3. The load voltage still follows the supply voltage.
- The DVR voltage is zero, because the DVR is off.

Figure 4.2 RMS voltages



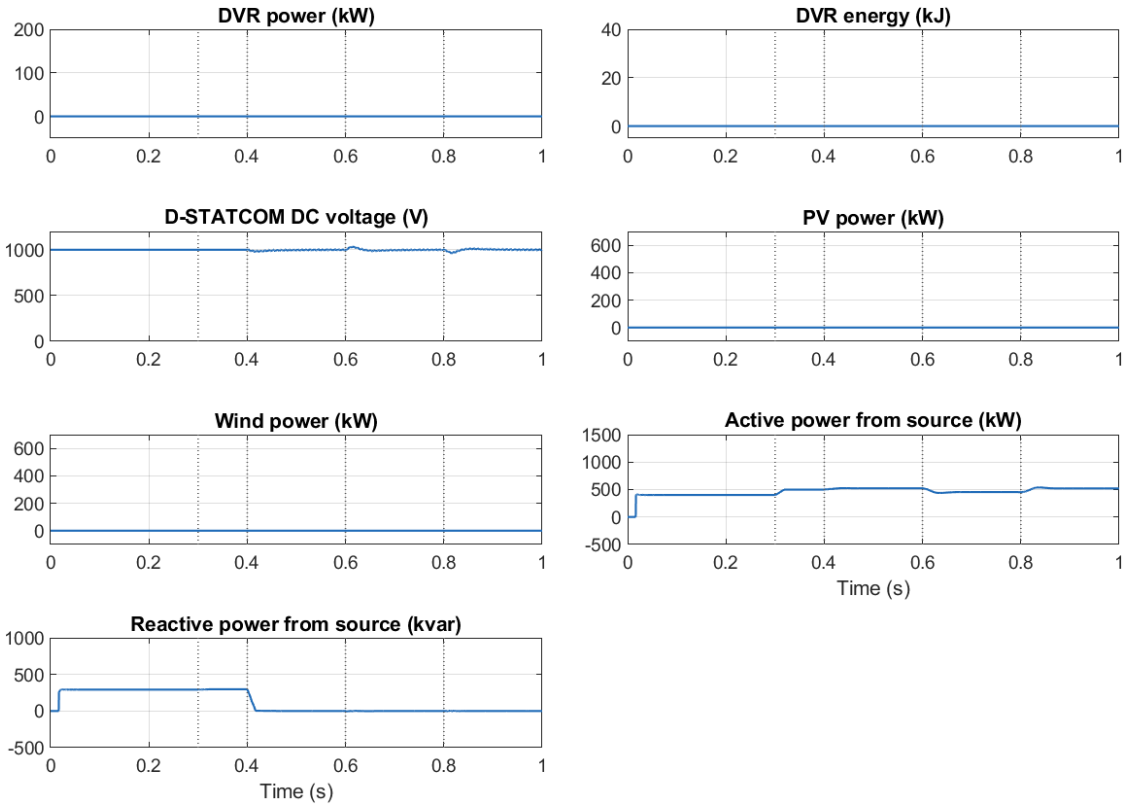
- At 0.4 s the load voltage rises from 0.989 / 1.015 / 0.991 to 1.008 / 1.031 / 1.005 pu, because reactive power no longer comes through the line.
- During the fault the load is 0.621 / 1.116 / 1.105 pu, only about 0.01 pu different from Step 3.

Figure 4.3 Power quality



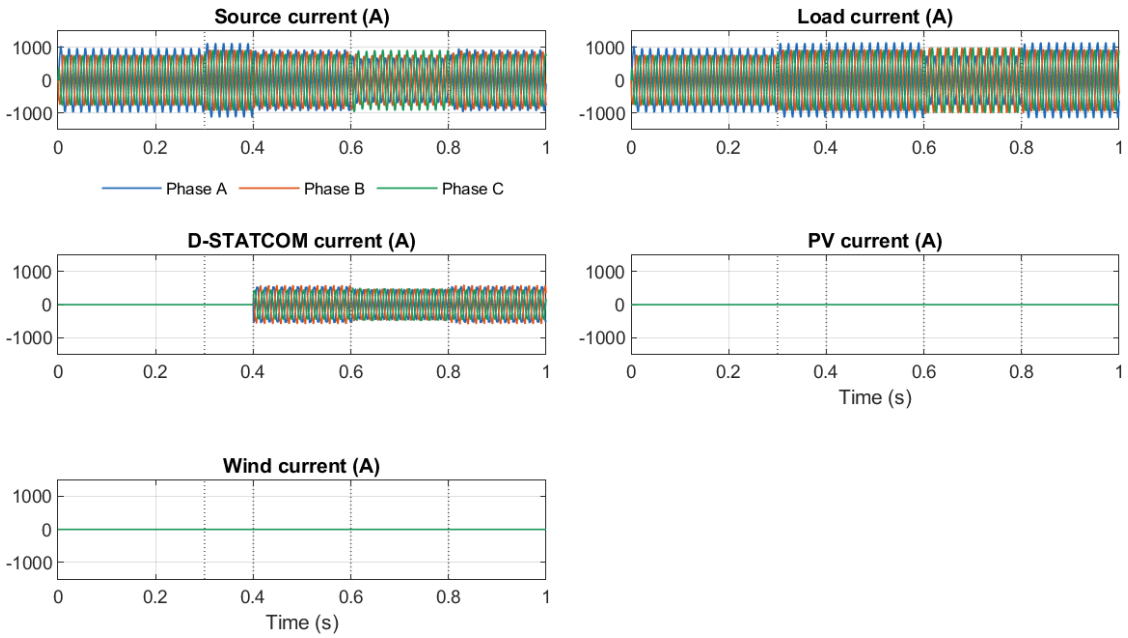
- This is the main result. At 0.4 s the power factor goes from 0.86 to 1.00.
- Current unbalance falls from 7.0 to 0.7 %, and current THD from about 5 to 2.8 %.
- Voltage unbalance still rises during the fault, to about 9 %.

Figure 4.4 Power and devices



- Reactive power from the source falls from 297 kvar to almost 0: the D-STATCOM now supplies it.
- Active power rises from 499 to 522 kW, because the load voltage is higher and the D-STATCOM has losses.
- The DC voltage stays close to 1000 V, between 958 and 1036 V after the switch-on.

Figure 4.5 Currents



- D-STATCOM current appears at 0.4 s, with peaks of about 460 to 580 A.
- The peak source current on phase A falls from 1133 to 938 A, while the load current stays about the same. The D-STATCOM supplies the difference.

Try the voltage regulation mode

Set **D-STATCOM mode** to 2 and run. Compare with Step 3, using the fault tick boxes for each row.

Fault	Without D-STATCOM (pu)	Mode 2 (pu)
LLLG	0.604 / 0.631 / 0.617	0.627 / 0.656 / 0.638
LG, phase A	0.609 / 1.100 / 1.091	0.631 / 1.128 / 1.116

It raises the voltage by only about 0.02 pu, and in the LG fault it also raises the swell. A shunt device of this size cannot restore a deep sag caused by an upstream fault, which is why Step 5 adds a series device.

Set the mode back to **1** and the tick boxes to **Phase A and Ground** before Step 5.

Step 5: DVR alone

Aim. See what a series device can do. The DVR adds a voltage in series with the line, so it can correct the voltage the load receives.

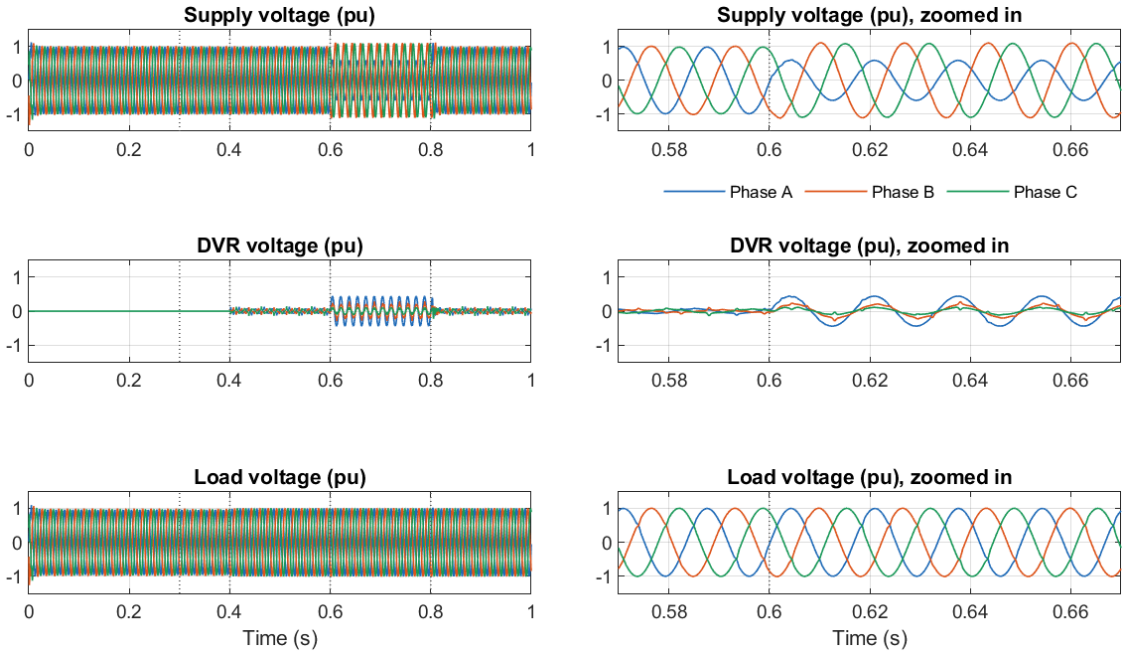
Change these blocks (inside Settings)

Block	Field	Set to
D-STATCOM in service (1 = yes, 0 = no)	Constant value	0
DVR in service (1 = yes, 0 = no)	Constant value	1
DVR gains [KpV; KiV; limit (pu)]	Constant value	[0.2;200;0.5]

These are the hand-tuned gains. Press **Run**. Dotted lines: rectifier on at 0.3 s, DVR on at 0.4 s, fault from 0.6 s to 0.8 s.

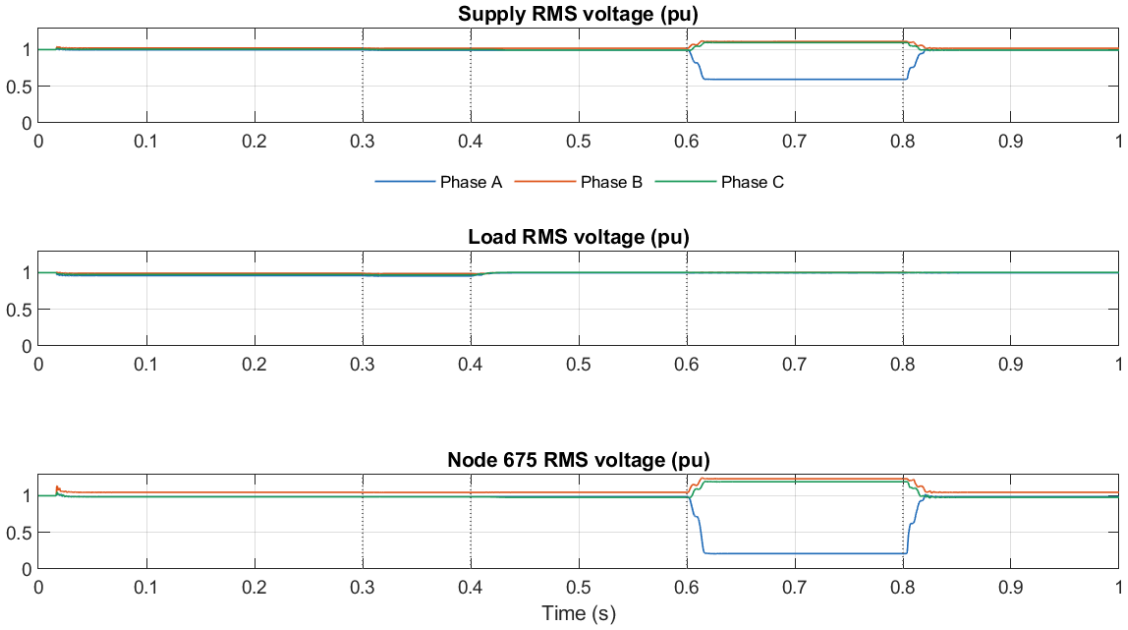
Result. The supply sags to 0.59 pu and swells to 1.11 pu, but the load stays at 1.00 pu. The DVR repairs the voltage, but it does not improve the power factor or the current.

Figure 5.1 Voltage waveforms



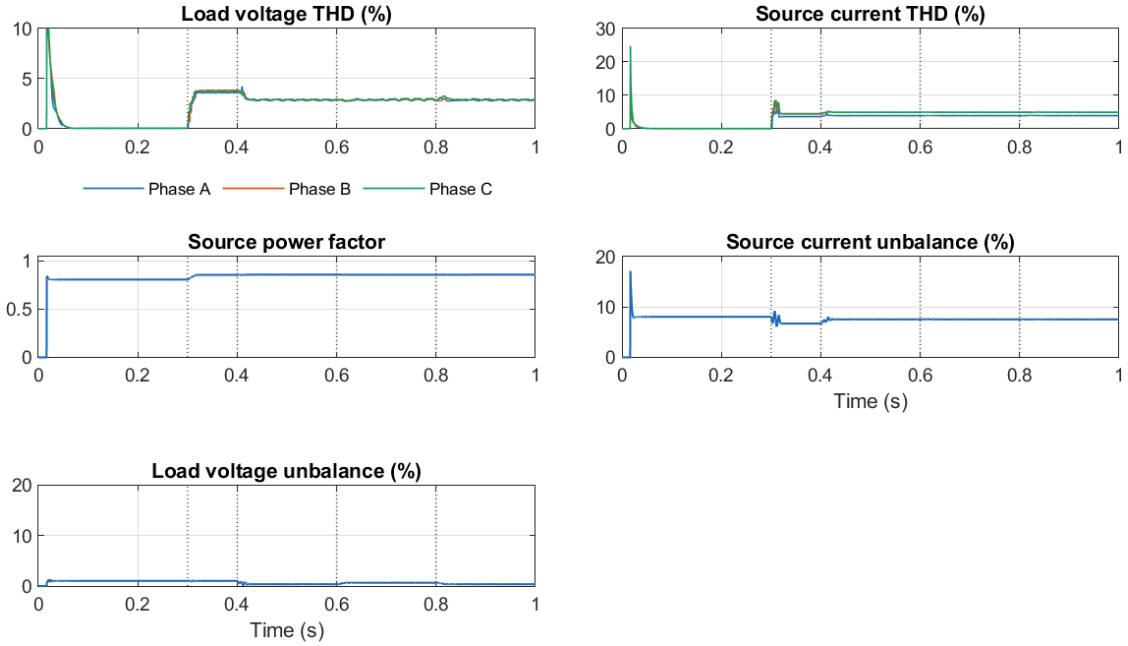
- This is the most important figure of this step. The supply voltage changes at 0.6 s, the DVR voltage appears, and the load voltage stays a clean sine wave.
- The DVR adds about 0.44 pu on phase A and less on phases B and C. Supply plus DVR voltage gives the load voltage.

Figure 5.2 RMS voltages



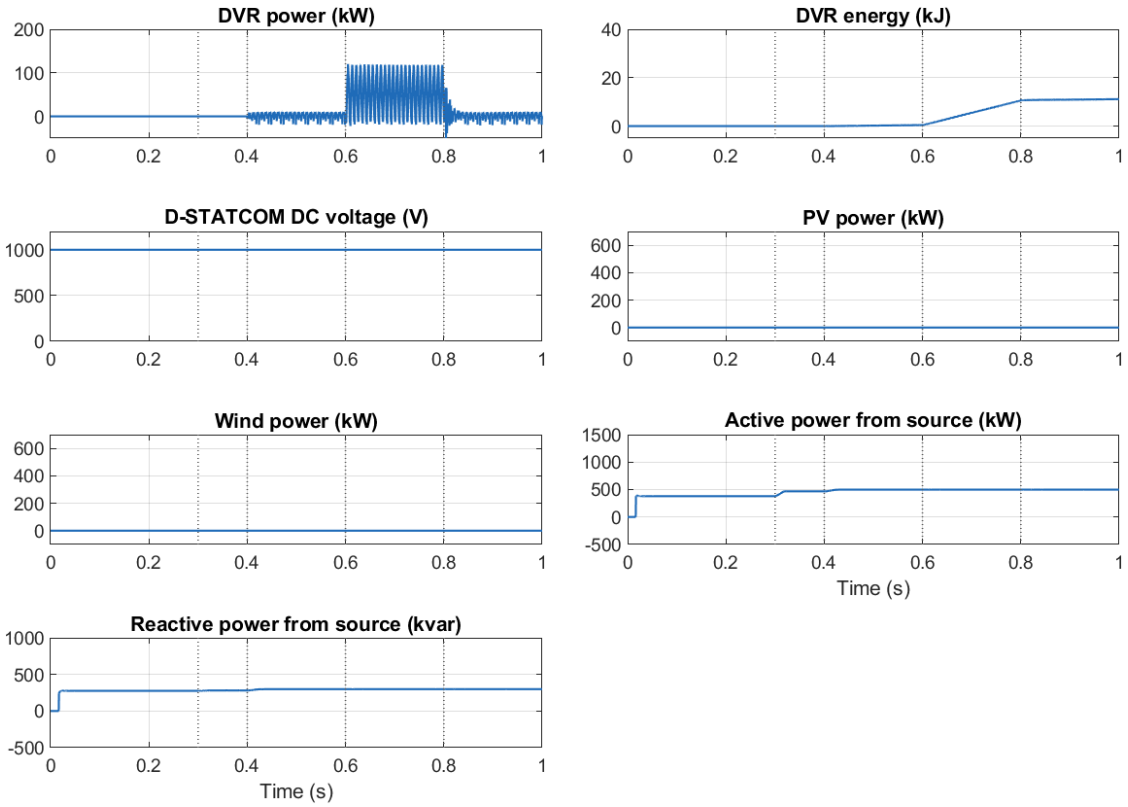
- Before 0.4 s the load is low, 0.954 / 0.986 / 0.969 pu. The DVR is in the line but not started, so its series impedance drops some voltage.
- At 0.4 s it starts, and the load rises to 0.997 / 1.002 / 1.002 pu.
- During the fault the supply is 0.594 / 1.109 / 1.095 pu and the load stays at 0.997 / 1.004 / 1.001 pu.
- Node 675 still falls to 0.21 pu. The DVR protects only the load behind it.

Figure 5.3 Power quality



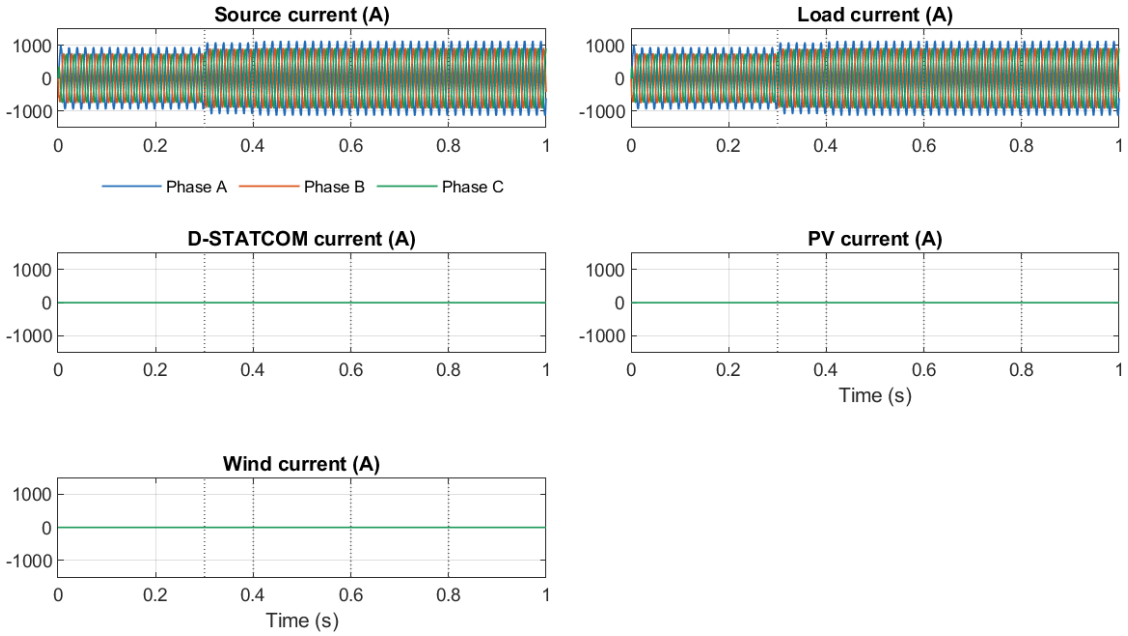
- Load voltage unbalance stays below 0.7 % during the fault, against 8.9 % in Step 3.
- The power factor stays at 0.86 and the current unbalance at 7.5 %. The DVR does not change the current.
- Load voltage THD is 2.9 %, higher than the 2 % of Step 3, because of the series impedance.

Figure 5.4 Power and devices



- DVR power swings up and down during the fault, between about −20 and 120 kW. Its average is about 51 kW. The display shows the peak, 121 kW.
- DVR energy rises in a straight line during the fault, by about 10 kJ. This energy must come from the DVR's storage.
- Power from the source does not change during the fault: the load does not notice the fault.

Figure 5.5 Currents



- The current stays the same before and during the fault, with peaks of 1142 / 921 / 921 A. Compare this with Figure 3.5, where it changes.

Try the other fault types

Change the tick boxes of **Fault 680** and run again. Values are during the fault; peak power and energy are from the displays.

Fault	Supply (pu)	Load (pu)	Peak DVR power (kW)	DVR energy (kJ)
LLLG	0.591 / 0.624 / 0.607	0.996 / 1.003 / 1.002	283	40.7
LG, phase A	0.594 / 1.109 / 1.095	0.997 / 1.004 / 1.001	121	11.1
LL, phases B, C	0.979 / 0.725 / 0.728	0.995 / 1.010 / 0.996	288	20.8
LLG, phases B, C	1.104 / 0.605 / 0.604	0.996 / 1.006 / 0.999	291	22.0

The load stays within about 1 % in all four faults. Set the tick boxes back to **Phase A and Ground** before Step 6.

Step 6: PV and wind plants

Aim. See what the generation does to the feeder, and whether the plants stay connected during a fault. The PV plant is 400 kW at node 634, and the wind plant is 500 kW at node 675.

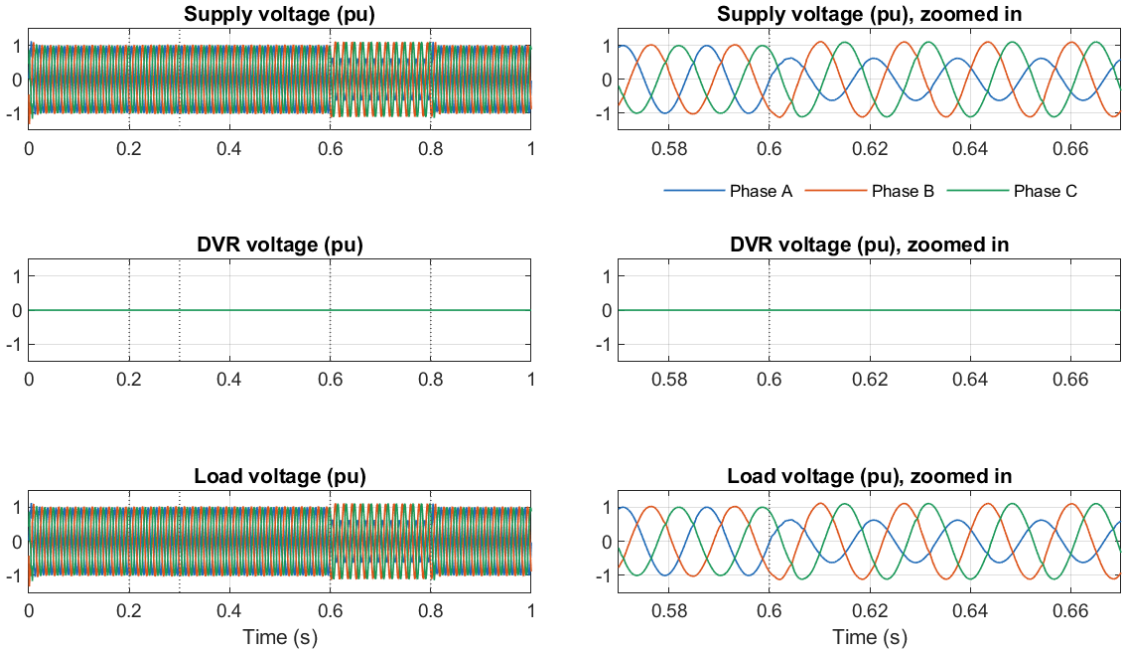
Change these blocks (inside Settings)

Block	Field	Set to
DVR in service (1 = yes, 0 = no)	Constant value	0
PV plant in service (1 = yes, 0 = no)	Constant value	1
Wind plant in service (1 = yes, 0 = no)	Constant value	1

Press **Run**. Dotted lines: PV and wind on at 0.2 s, rectifier on at 0.3 s, fault from 0.6 s to 0.8 s.

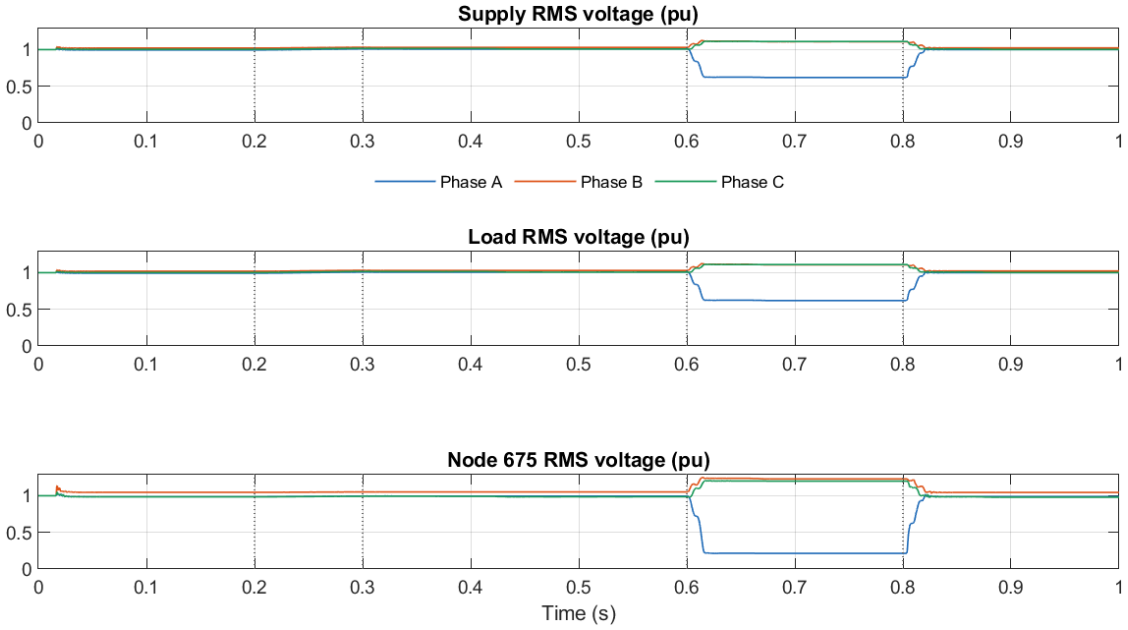
Result. The plants give about 390 kW and 495 kW and keep running in this fault. They barely lift the sag, from 0.61 to 0.62 pu, so the load still needs the DVR. The power factor at node 634 gets worse.

Figure 6.1 Voltage waveforms



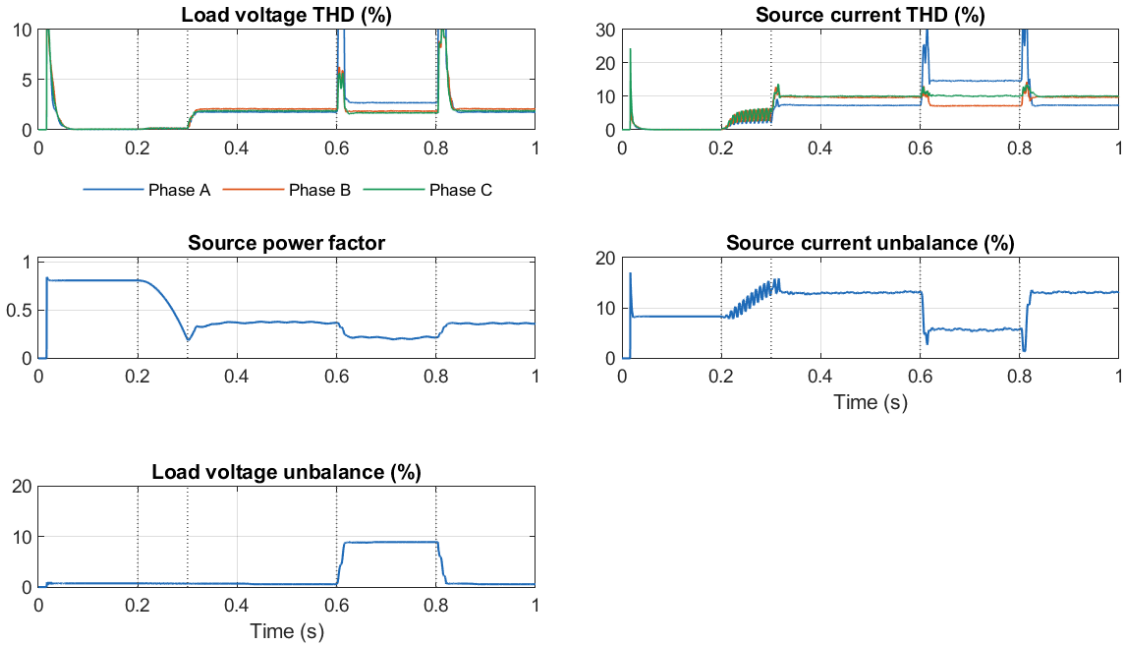
- The fault looks the same as in Step 3: the load still sags and swells. The DVR voltage is zero.

Figure 6.2 RMS voltages



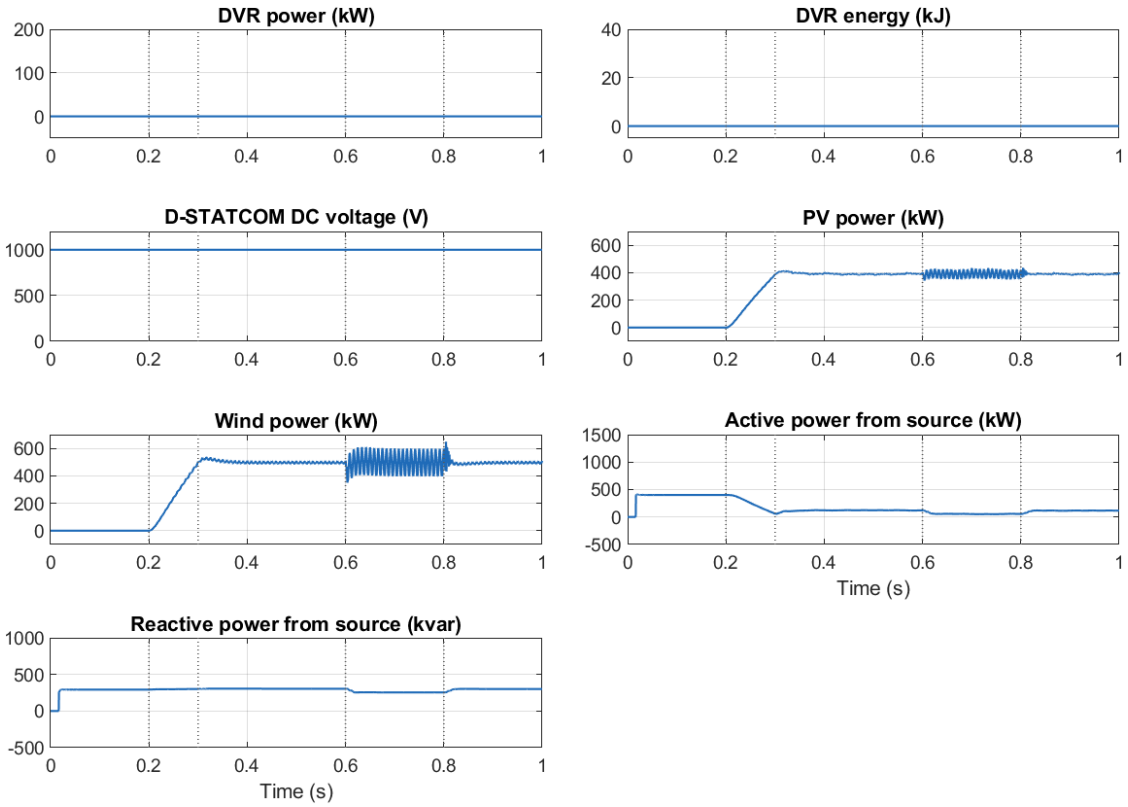
- After 0.2 s the voltage rises by about 0.01 pu, because less power flows through the feeder.
- During the fault the load is 0.620 / 1.107 / 1.110 pu, almost the same as in Step 3.
- Node 675 falls to 0.213 / 1.229 / 1.199 pu.

Figure 6.3 Power quality



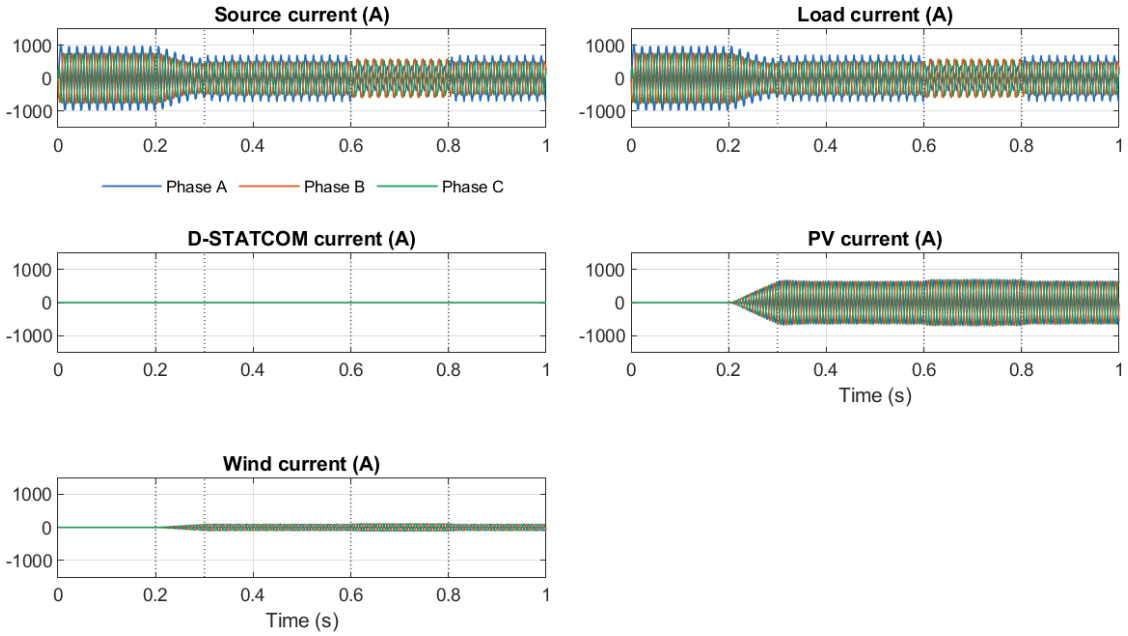
- The power factor falls to about 0.38. This looks bad, but the PV plant is not at fault. It now supplies most of the active power, so the source gives little active power but still all of the reactive power.
- Current THD rises to about 10 % and current unbalance to about 13 %. The source current is now smaller, so the same harmonics and unbalance are a larger share of it.

Figure 6.4 Power and devices



- Both plants ramp up over 0.1 s after 0.2 s, because of the soft start.
- Active power from the source falls from 497 to 125 kW. This fall is about the PV power only: the wind plant is at node 675, so its power does not pass through this measurement.
- During this LG fault the plants' power swings at twice line frequency, because the voltage is unbalanced. The averages stay about the same: PV 389 kW and wind 494 kW.

Figure 6.5 Currents



- PV current peaks at about 665 A, and wind current at only about 96 A. The wind plant works at 4.16 kV, so it needs much less current for its power.
- The source current falls to 728 / 544 / 534 A. Load current is the same as source current here, because PV current is already taken out of it.

Try a three-phase fault

Tick A, B, C and Ground in **Fault 680** and run.

Fault	Load (pu)	Node 675 (pu)	PV during fault	Wind during fault
LG, phase A	0.620 / 1.107 / 1.110	0.213 / 1.229 / 1.199	389 kW	494 kW
LLLG	0.617 / 0.641 / 0.633	0.207 / 0.219 / 0.212	307 kW, at its current limit	0 kW, ceased

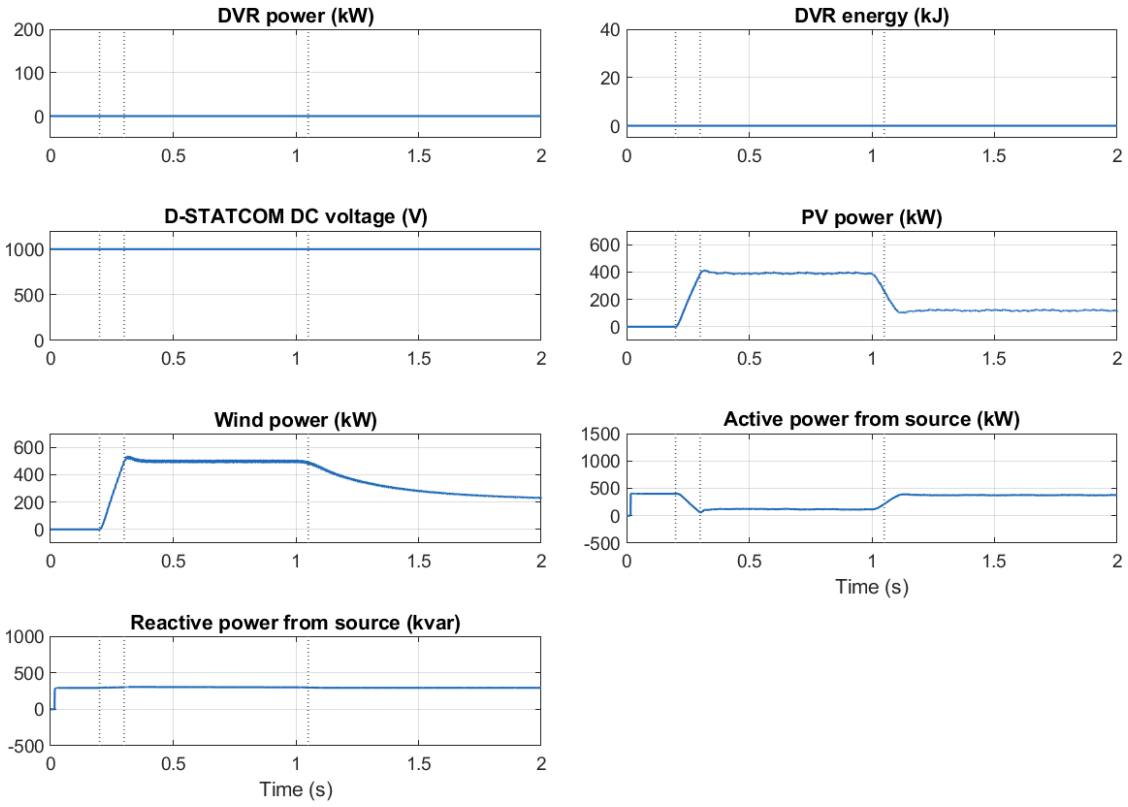
In the three-phase fault node 675 falls below 0.5 pu on all phases, so the wind plant stops injecting. It returns after the fault.

Set the tick boxes back to **Phase A and Ground**.

Try less sun and wind

Block	Field	Set to
Sun: irradiance (W/m2) against time (s)	Breakpoints / Table data	[0 1.0 1.1 100] / [1000 1000 300 300]
Wind speed (m/s) against time (s)	Breakpoints / Table data	[0 1.0 1.1 100] / [12 12 9 9]
Fault 680	Switching times (s)	[100 100.1]
Stop time (Simulation tab)		2

Figure 6.6 Power and devices with less sun and wind



- The dotted line near 1.05 s marks the change, which runs from 1.0 s to 1.1 s.
- PV power falls from 396 to 119 kW quickly. Wind power falls from 510 kW, reaching 231 kW at 2 s and still falling, because the turbine slows gradually.
- Active power from the source rises from 114 to 378 kW to make up the loss.
- The load voltage moves by only about 0.01 pu. On this feeder a change of sun and wind is a mild disturbance.

Before Step 7, set these back: Sun Breakpoints [0 100], Table data [1000 1000]; Wind speed Breakpoints [0 100], Table data [12 12]; Fault 680 times [0.6 0.8]; stop time 1.

Step 7: everything together

Aim. Run all four devices together with the optimised (PSO) gains. These are the saved values of the model.

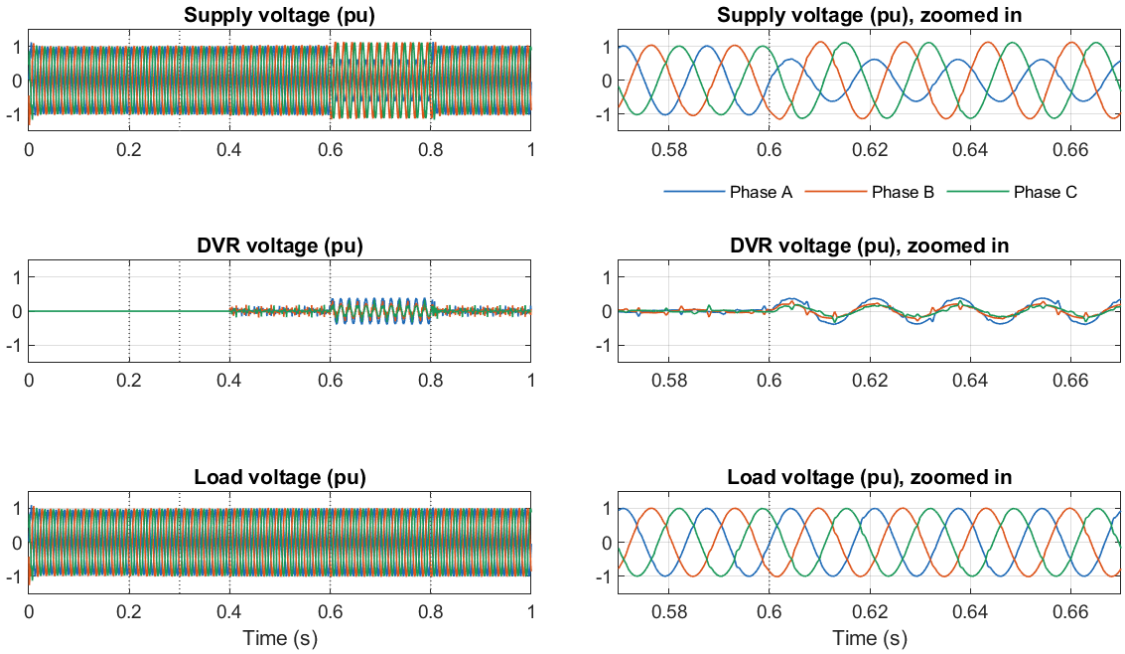
Change these blocks (inside Settings)

Block	Field	Set to
DVR in service (1 = yes, 0 = no)	Constant value	1
D-STATCOM in service (1 = yes, 0 = no)	Constant value	1
DVR gains [KpV; KiV; limit (pu)]	Constant value	[0.9;20;0.5]
D-STATCOM gains [bandwidth (Hz); KpDc; KiDc; KpV; KiV; Imax (A)]	Constant value	[1633.46;4.06058;34.8962;5;500;850]

PV and wind stay at 1. Press **Run**. Dotted lines: PV and wind on at 0.2 s, rectifier on at 0.3 s, DVR and D-STATCOM on at 0.4 s, fault from 0.6 s to 0.8 s.

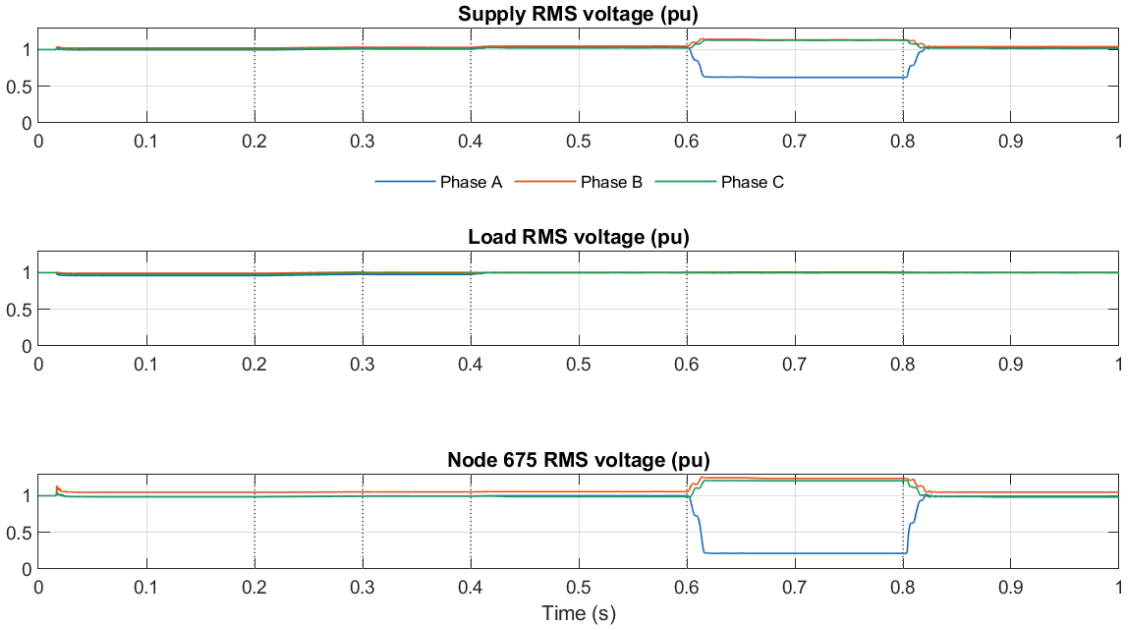
Result. The load stays within 1 % through the fault, the power factor is 1.00, and the DVR needs much less power: 47 kW peak instead of 121 kW.

Figure 7.1 Voltage waveforms



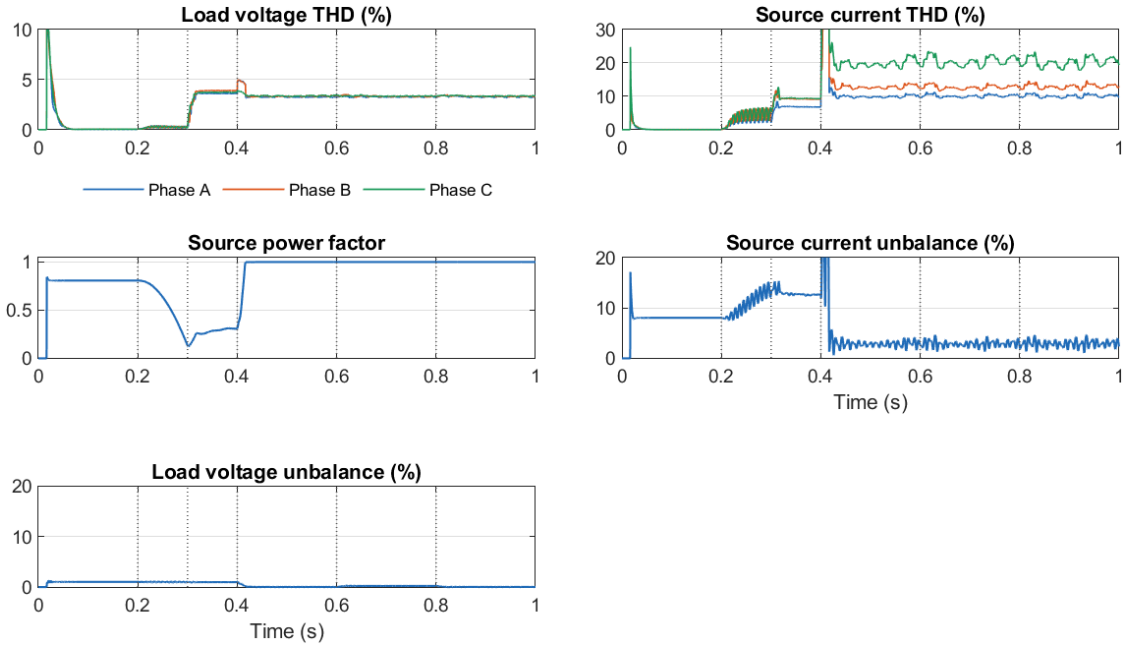
- The same picture as Step 5: the supply changes at 0.6 s, the DVR adds its voltage, and the load stays a clean sine wave.

Figure 7.2 RMS voltages



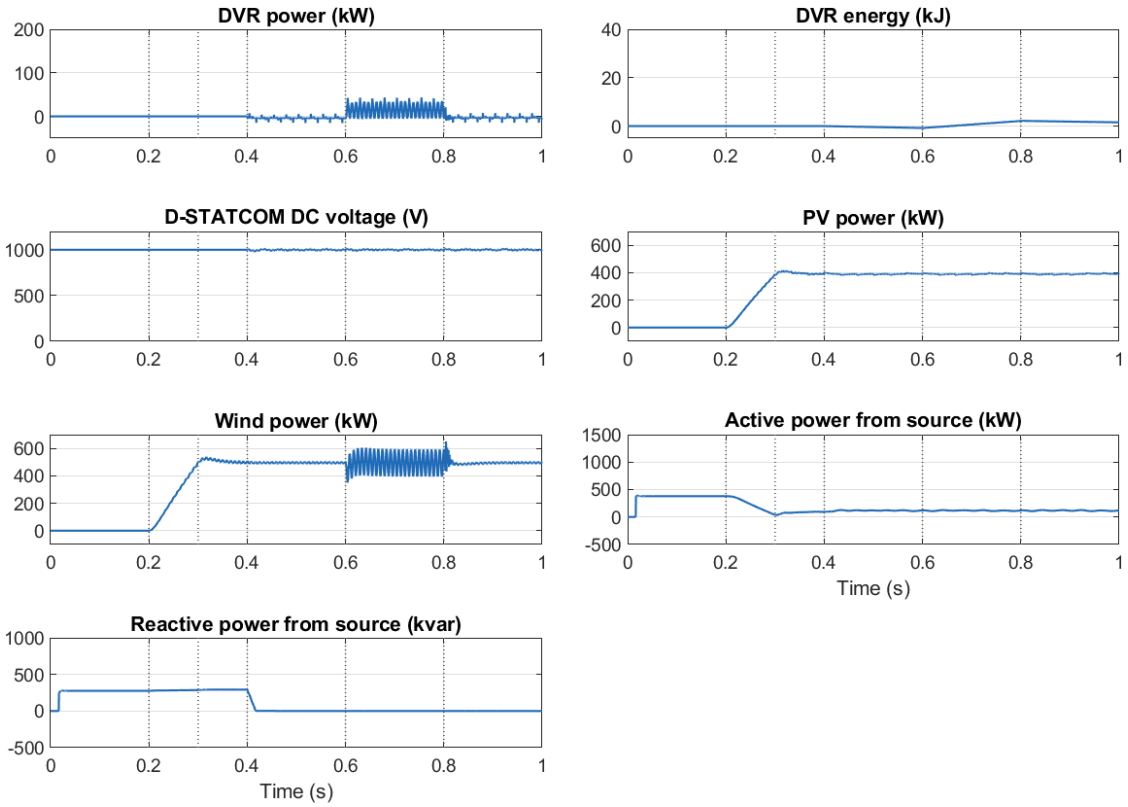
- Before 0.4 s the load is 0.974 / 1.003 / 0.988 pu. After the devices start it is 0.998 / 1.004 / 0.998 pu.
- During the fault the supply is 0.621 / 1.132 / 1.126 pu and the load stays at 0.998 / 1.007 / 0.997 pu.
- Node 675 still falls to 0.213 pu, because it is outside the DVR's protection.

Figure 7.3 Power quality



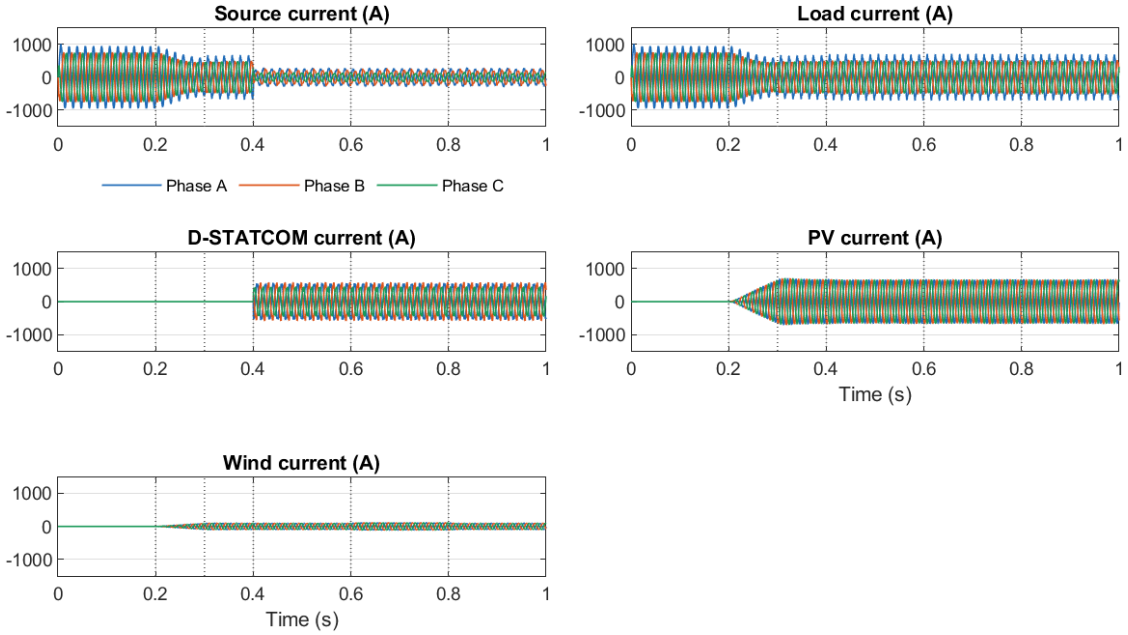
- The power factor jumps from 0.31 to 1.00 at 0.4 s, and load voltage unbalance falls to about 0.02 %.
- Load voltage THD is about 3.3 %.
- Current THD reads 10 to 22 %, which looks high. It is high only because the source current is now small. Against the 500 kVA transformer rating the distortion is about 3 %.

Figure 7.4 Power and devices



- During the fault DVR power averages about 15 kW, against 51 kW in Step 5. The PV plant and the D-STATCOM behind the DVR supply most of the load, so less power passes through it.
- DVR energy is slightly negative before the fault. The supply is above 1 pu there, so the DVR lowers the voltage and absorbs a little power.
- Reactive power from the source is almost 0 and active power about 117 kW. The D-STATCOM DC voltage stays between 984 and 1014 V.

Figure 7.5 Currents



- The peak source current on phase A is only about 283 A, against 1133 A in Step 3. The devices supply most of what the load needs.
- D-STATCOM current peaks at about 455 to 585 A, PV at about 670 A, and wind at about 96 A.

All steps side by side

All with the saved fault: phase A to ground, 0.6 s to 0.8 s. Read these from the displays.

Display	Step 3 Base	Step 4 D-STATCOM	Step 5 DVR	Step 6 PV, wind	Step 7 All
Lowest load voltage (pu)	0.608	0.620	0.996	0.619	0.996
Highest load voltage (pu)	1.11	1.12	1.00	1.12	1.01
Power factor at the end	0.86	1.00	0.86	0.36	1.00
Current unbalance at the end (%)	7.1	0.7	7.5	13.2	2.3
Source current THD at the end, worst phase (%)	5.1	2.8	4.9	10.1	19.3
Peak DVR power (kW)	0	0	121	0	46.6

Compare the hand-tuned and PSO gains

Run all four faults with each gain set. The load voltage is held within about 1 % with both; the main gains are lower voltage THD and lower current unbalance.

Gains	Load voltage THD at the end (%)	Current unbalance at the end (%)	Peak DVR power, LLLG / LG / LL / LLG (kW)
Hand-tuned	3.7	2.9	64 / 41 / 62 / 64
PSO (saved)	3.3	2.3–2.4	66 / 47 / 65 / 66

Find the DVR size

The third number in **DVR gains** is the injection limit. Use the hand-tuned gains and an LLLG fault, and change only this number.

Injection limit (pu)	Lowest load voltage (pu)	Result
0.2	0.857 / 0.889 / 0.872	Load below 0.9 pu
0.3	0.949 / 0.960 / 0.958	Load above 0.9 pu
0.4	0.990 / 0.988 / 0.993	Load restored
0.5	0.995 / 0.997 / 0.995	Load restored

Find the D-STATCOM size

The last number in **D-STATCOM gains** is the current limit Imax. Use the hand-tuned gains and no fault, and change only this number.

Imax (A peak)	Power factor at the end	Current unbalance at the end (%)
200	0.52	16.9
400	0.87	20.2
600	1.00	3.9
850	1.00	2.9

A limit of 600 A, about 350 kVA, is enough for a power factor of 1.00. The 500 kVA assumed at first is more than needed.

What Step 7 cannot do by hand. Finding the PSO gains needs 56 simulations for each optimiser run, so that search was done by a script, and its answer is saved in this model. Putting the D-STATCOM on the supply side of the DVR is a different circuit, in the separate model `IEEE13_full_supply.slx`.

Reference: every value you can change

Block	Field to edit	Saved value	What it means
Inside Settings			
DVR in service (1 = yes, 0 = no)	Constant value	1	0 takes the DVR out: a bypass breaker closes around it
DVR switch-on time (s)	Step time	0.4	When the DVR starts injecting
DVR gains [KpV; KiV; limit (pu)]	Constant value	[0.9;20;0.5]	PSO gains. Hand-tuned: [0.2;200;0.5]
D-STATCOM in service (1 = yes, 0 = no)	Constant value	1	0 keeps its breaker open
D-STATCOM switch-on time (s)	Step time	0.4	When its breaker closes and control starts
D-STATCOM mode (1 = load compensation, 2 = voltage regulation)	Constant value	1	1 corrects the load current. 2 supports the voltage
D-STATCOM gains [bandwidth (Hz); KpDc; KiDc; KpV; KiV; Imax (A)]	Constant value	[1633.46; 4.06058; 34.8962; 5; 500; 850]	PSO gains. Hand-tuned: [1500;1;15;5;500;850]
PV plant in service (1 = yes, 0 = no)	Constant value	1	0 keeps its breaker open
PV plant switch-on time (s)	Step time	0.2	When the PV plant connects
Sun: irradiance (W/m2) against time (s)	Breakpoints and Table data	[0 100] and [1000 1000]	Irradiance at each time
Wind plant in service (1 = yes, 0 = no)	Constant value	1	0 keeps its breaker open
Wind plant switch-on time (s)	Step time	0.2	When the wind plant connects
Wind speed (m/s) against time (s)	Breakpoints and Table data	[0 100] and [12 12]	Wind speed at each time
Lowest and highest values taken from (s)	Constant value	0.5	The lowest and highest displays count from this time
On the main page			
Fault 680	Phase A, B, C, Ground tick boxes	A and Ground	Fault type
Fault 680	Switching times (s)	[0.6 0.8]	[start end] of the fault
Fault 680	Fault resistance Ron (Ohm)	0.01	A larger value gives a milder fault
NL Breaker	Switching times (s)	[0.3]	When the rectifier load connects
Toolbar			
Stop time (Simulation tab)		1.0	Length of the run in seconds

Good to know

- Change values only inside **Settings**, in **Fault 680** and in **NL Breaker**. Leave the device subsystems and the blocks inside **Results** as they are.
- Keep the switch-on times before the fault, and the 0.5 s start of the lowest and highest values after them, so that start-up does not count.
- In the Sun and Wind speed tables, the times must increase, and both rows must have the same number of values.
- MATLAB may print *“Matrix is close to singular or badly scaled”*. The original IEEE 13-bus model prints the same warning, and it does not stop the run.
- Simulink makes a folder `slprj` and a `.slxc` file next to the model. They are temporary and can be deleted.
- If Simulink reports missing blocks, Simscape Electrical is not installed.